

LangChain serialization injection vulnerability enables secret extraction in dumps/loads APIs

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Content

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LangChain serialization injection vulnerability enables secret extraction in dumps/loads APIs

- Advisory: GHSA-c67j-w6g6-q2cm
- CVE: CVE-2025-68664
- Severity: critical
- Published: 2025-12-23
- Updated: 2025-12-24

Affected

- `langchain-core` (pip): $\geq 1.0.0$, $< 1.2.5$, fixed in 1.2.5
- `langchain-core` (pip): $< 0.3.81$, fixed in 0.3.81

Description

Summary

A serialization injection vulnerability exists in LangChain's `dumps()` and `dumpd()` functions. The functions do not escape dictionaries with `'lc'` keys when serializing free-form dictionaries. The `'lc'` key is used internally by LangChain to mark serialized objects. When user-controlled data contains this key structure, it is treated as a legitimate LangChain object during deserialization rather than plain user data.

Attack surface

The core vulnerability was in `dumps()` and `dumpd()`: these functions failed to escape user-controlled dictionaries containing `'lc'` keys. When this unescaped data was later deserialized via `load()` or `loads()`, the injected structures were treated as legitimate LangChain objects rather than plain user data.

This escaping bug enabled several attack vectors:

1. **Injection via user data:** Malicious LangChain object structures could be injected through user-controlled fields like `metadata`, `additional_kwargs`, or `response_metadata`
2. **Class instantiation within trusted namespaces:** Injected manifests could instantiate any `Serializable` subclass, but only within the pre-approved trusted namespaces (`langchain_core`, `langchain`, `langchain_community`). This includes classes with side effects in `__init__` (network calls, file operations, etc.). Note that namespace validation was already enforced before this patch, so arbitrary classes outside these trusted namespaces could not be instantiated.

Security hardening

This patch fixes the escaping bug in `dumps()` and `dumpd()` and introduces new restrictive defaults in `load()` and `loads()`: allowlist enforcement via `allowed_objects="core"` (restricted to [serialization mappings](#)), `secrets_from_env` changed from `True` to `False`, and default Jinja2 template blocking via `init_validator`. These are breaking changes for some use cases.

Who is affected?

Applications are vulnerable if they:

1. **Use `astream_events(version="v1")`** — The v1 implementation internally uses vulnerable serialization. Note: `astream_events(version="v2")` is not vulnerable.
2. **Use `Runnable.astream_log()`** — This method internally uses vulnerable serialization for streaming outputs.
3. **Call `dumps()` or `dumpd()` on untrusted data, then deserialize with `load()` or `loads()`** — Trusting your own serialization output makes you vulnerable if user-controlled data (e.g., from LLM responses, metadata fields, or user inputs) contains `'lc'` key structures.
4. **Deserialize untrusted data with `load()` or `loads()`** — Directly deserializing untrusted data that may contain injected `'lc'` structures.
5. **Use `RunnableWithMessageHistory`** — Internal serialization in message history handling.

6. Use `InMemoryVectorStore.load()` to deserialize untrusted documents.
7. Load untrusted generations from cache using `langchain-community caches`.
8. Load untrusted manifests from the LangChain Hub via `hub.pull`.
9. Use `StringRunEvaluatorChain` on untrusted runs.
10. Use `create_lc_store` or `create_kv_docstore` with untrusted documents.
11. Use `MultiVectorRetriever` with byte stores containing untrusted documents.
12. Use `LangSmithRunChatLoader` with runs containing untrusted messages.

The most common attack vector is through **LLM response fields** like `additional_kwargs` or `response_metadata`, which can be controlled via prompt injection and then serialized/deserialized in streaming operations.

Impact

Attackers who control serialized data can extract environment variable secrets by injecting `{"lc": 1, "type": "secret", "id": ["ENV_VAR"]}` to load environment variables during deserialization (when `secrets_from_env=True`, which was the old default). They can also instantiate classes with controlled parameters by injecting constructor structures to instantiate any class within trusted namespaces with attacker-controlled parameters, potentially triggering side effects such as network calls or file operations.

Key severity factors:

- Affects the serialization path - applications trusting their own serialization output are vulnerable
- Enables secret extraction when combined with `secrets_from_env=True` (the old default)
- LLM responses in `additional_kwargs` can be controlled via prompt injection

Exploit example

```

from langchain_core.load import dumps, load
import os

# Attacker injects secret structure into user-controlled data
attacker_dict = {
    "user_data": {
        "lc": 1,
        "type": "secret",
        "id": ["OPENAI_API_KEY"]
    }
}

serialized = dumps(attacker_dict) # Bug: does NOT escape the 'lc' key

os.environ["OPENAI_API_KEY"] = "sk-secret-key-12345"
deserialized = load(serialized, secrets_from_env=True)

print(deserialized["user_data"]) # "sk-secret-key-12345" - SECRET LEAKED!

```

Security hardening changes (breaking changes)

This patch introduces three breaking changes to `load()` and `loads()`:

1. **New `allowed_objects` parameter** (defaults to `'core'`): Enforces allowlist of classes that can be deserialized. The `'all'` option corresponds to the list of objects [specified in `mappings.py`](#) while the `'core'` option limits to objects within `langchain_core`. We recommend that users explicitly specify which objects they want to allow for serialization/deserialization.
2. **`secrets_from_env` default changed from `True` to `False`**: Disables automatic secret loading from environment
3. **New `init_validator` parameter** (defaults to `default_init_validator`): Blocks Jinja2 templates by default

Migration guide

No changes needed for most users

If you're deserializing standard LangChain types (messages, documents, prompts, trusted partner integrations like `ChatOpenAI`, `ChatAnthropic`, etc.), your code will work without changes:

```
from langchain_core.load import load
```

```
# Uses default allowlist from serialization mappings  
obj = load(serialized_data)
```

For custom classes

If you're deserializing custom classes not in the serialization mappings, add them to the allowlist:

```
from langchain_core.load import load  
from my_package import MyCustomClass
```

```
# Specify the classes you need  
obj = load(serialized_data, allowed_objects=[MyCustomClass])
```

For Jinja2 templates

Jinja2 templates are now blocked by default because they can execute arbitrary code. If you need Jinja2 templates, pass `init_validator=None`:

```
from langchain_core.load import load  
from langchain_core.prompts import PromptTemplate  
  
obj = load(  
    serialized_data,  
    allowed_objects=[PromptTemplate],  
    init_validator=None  
)
```

[!WARNING] Only disable `init_validator` if you trust the serialized data. Jinja2 templates can execute arbitrary Python code.

For secrets from environment

`secrets_from_env` now defaults to `False`. If you need to load secrets from environment variables:

```
from langchain_core.load import load  
  
obj = load(serialized_data, secrets_from_env=True)
```

Credits

- Dumps bug was reported by @yardenporat
- Changes for security hardening due to findings from @0xn3va and @VladimirEliTokarev

References

- <<https://github.com/langchain-ai/langchain/security/advisories/GHSA-c67j-w6g6-q2cm>>
- <<https://github.com/langchain-ai/langchain/pull/34455>>
- <<https://github.com/langchain-ai/langchain/pull/34458>>
- <<https://github.com/langchain-ai/langchain/commit/5ec0fa69de31bbe3d76e4cf9cd65a6accb8466c8>>
- <<https://github.com/langchain-ai/langchain/commit/d9ec4c5cc78960abd37da79b0250f5642e6f0ce6>>
- <<https://github.com/langchain-ai/langchain/releases/tag/langchain-core%3D%3D0.3.81>>
- <<https://github.com/langchain-ai/langchain/releases/tag/langchain-core%3D%3D1.2.5>>
- <<https://nvd.nist.gov/vuln/detail/CVE-2025-68664>>
- <<https://github.com/advisories/GHSA-c67j-w6g6-q2cm>>

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